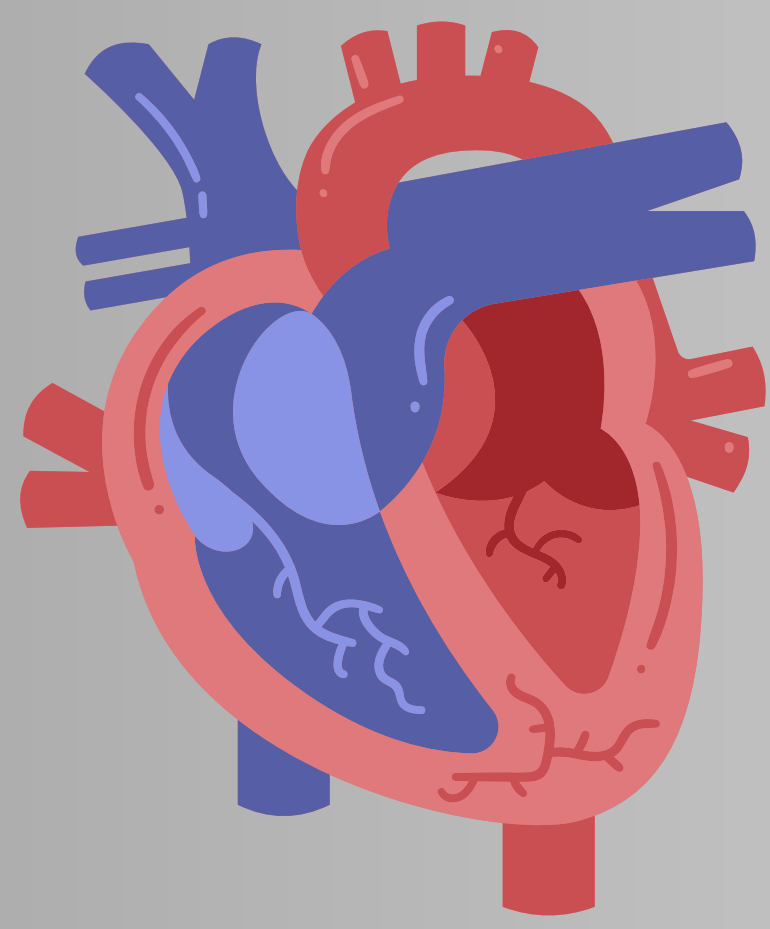




BLOOD BANK MANAGEMENT

PRANAV SAIRAM
RA2311026010195
V T S MUKUNDAN
RA2311026010209

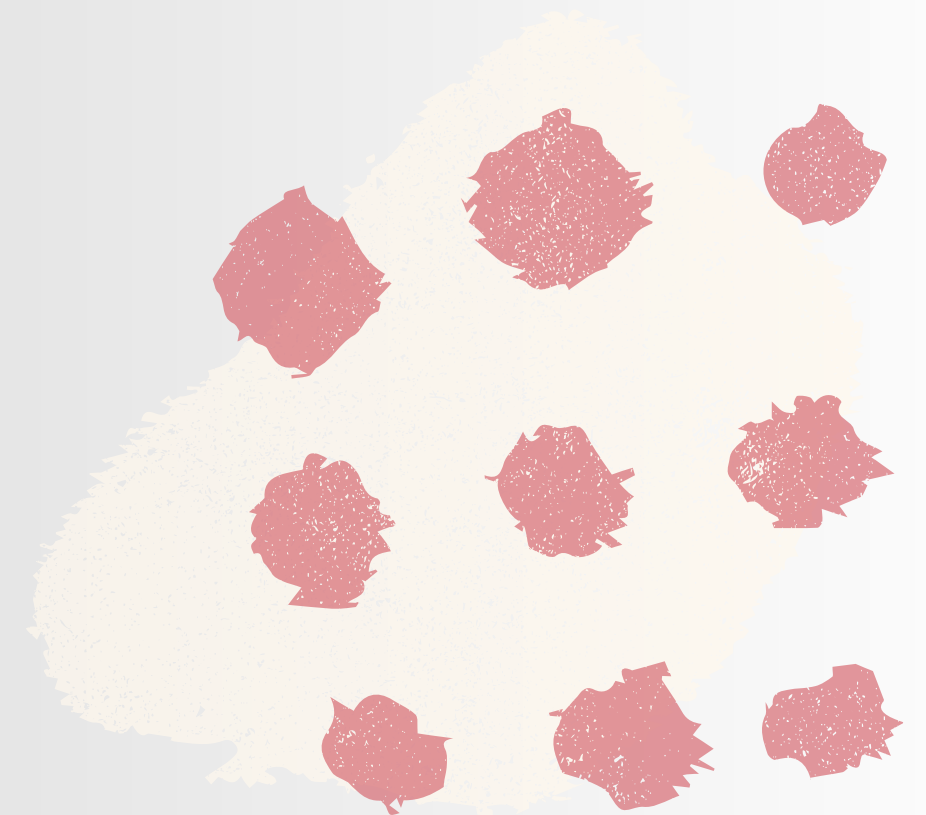
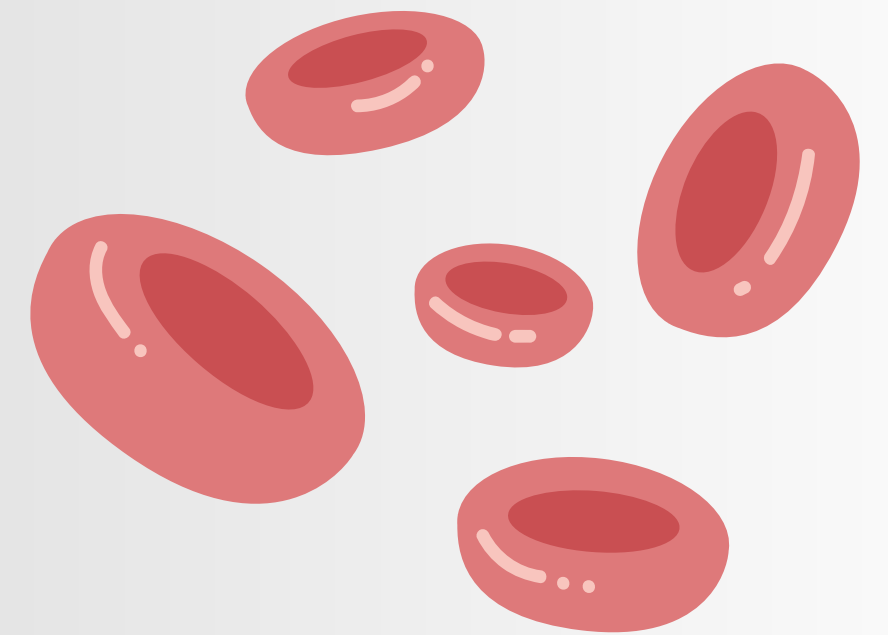


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INTRODUCTION

Blood banks are essential for collecting, storing, and distributing blood to patients in need. They classify donated blood by type, ensure its quality, and manage inventory. Traditional manual systems are time-consuming and inefficient in tracking blood quality and donor information. The Blood Bank Management System aims to automate these processes, providing an effective, time-efficient, and cost-effective solution that reduces manpower requirements.

PROBLEM STATEMENT

Current blood management relies heavily on manual processes, leading to delays in finding suitable blood donors or supplies, especially in emergencies. This system seeks to eliminate these inefficiencies by providing a centralized database for easy access to donor and blood availability information.

ABSTRACT

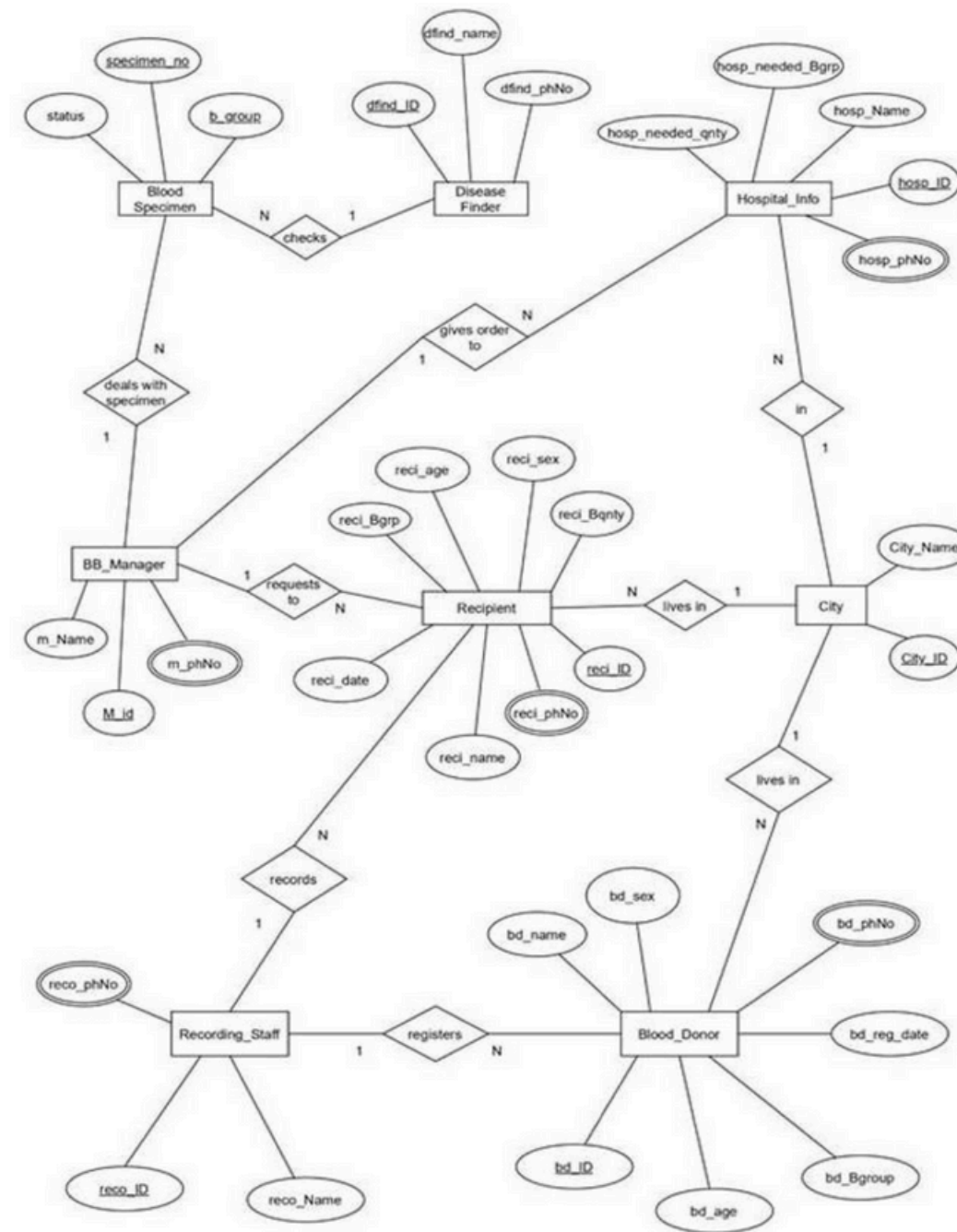
This project aims to develop a Blood Bank Management System that can be utilized by clinics, hospitals, laboratories, or during emergencies requiring blood units. The system facilitates the quick identification of the required blood types from both blood banks and donors, addressing the inefficiencies in current grapevine communication methods. The goal is to mitigate panic during emergencies by ensuring the timely availability of blood.



ER DIAGRAM

6

ER DIAGRAM

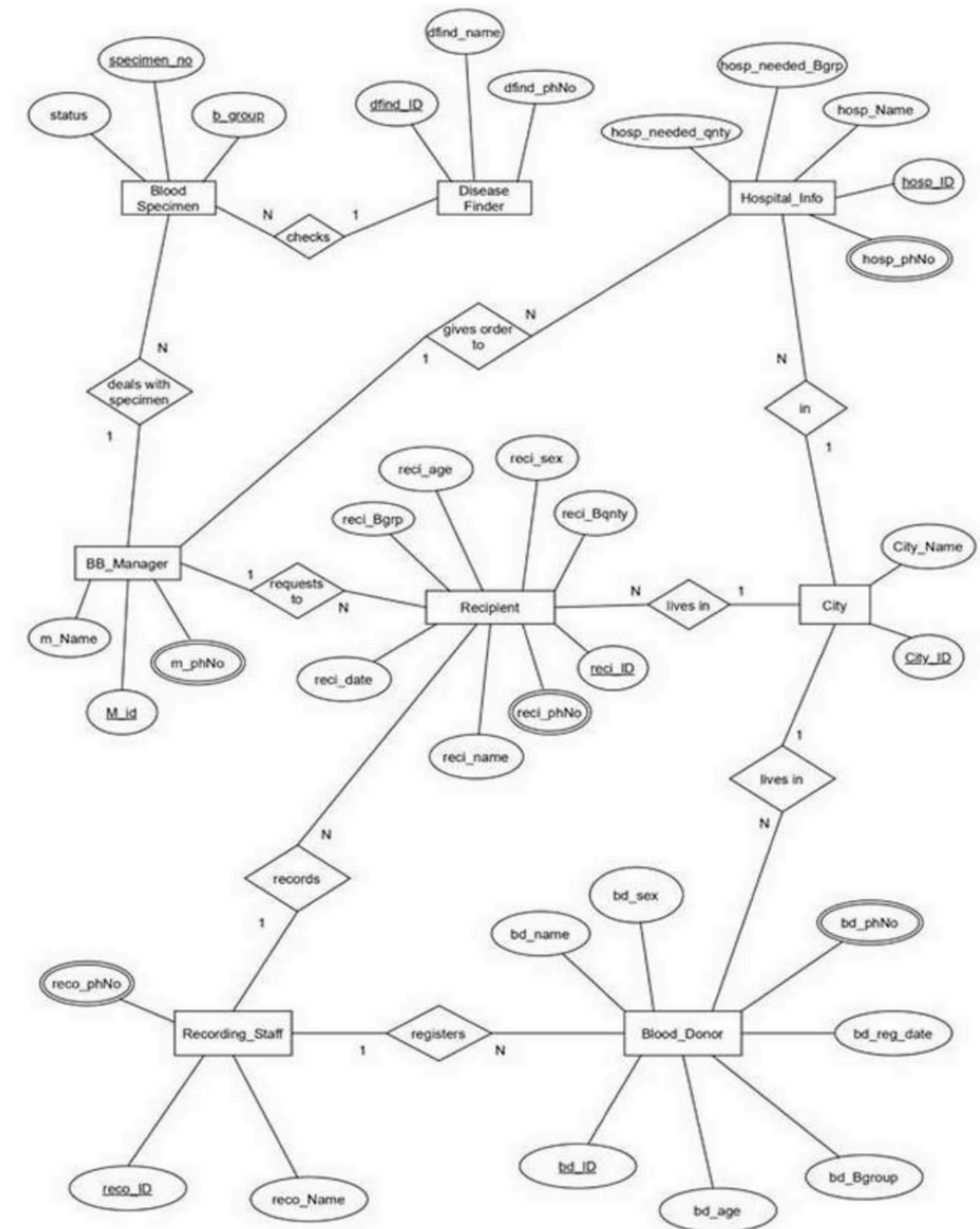


KEY ENTITIES & RELATIONSHIPS:

7

1. Blood Donor → Registers with Recording Staff, lives in a City.
2. Blood Specimen → Checked by Disease Finder, managed by BB Manager.
3. Recipient → Requests blood from BB Manager, lives in a City.
4. BB Manager → Handles blood requests and manages specimens.
5. Recording Staff → Registers donors and records recipient details.
6. Hospital Info → Requests blood, linked to a City.
7. Disease Finder → Checks blood samples.
8. City → Stores donor and recipient locations.

ER DIAGRAM



MODULES USED

The Modules used in the project

1. UI Frameworks & Libraries

- HTML, CSS, JavaScript – Basic front-end structure
- Bootstrap/Tailwind CSS – Responsive design
- React.js / Angular / Vue.js – Modern UI frameworks

2. Authentication Module

- Firebase Auth / OAuth / JWT – User login & authentication

3. API Handling Module

- Axios / Fetch API – Communication with the backend

4. Form Handling & Validation

- React Hook Form / Formik / Yup – User input validation

5. State Management

- Redux (React) / Vuex (Vue.js) / NgRx (Angular) – Manage application state

6. UI Components & Charts

- Material-UI / Ant Design / Chakra UI – Prebuilt UI components
- Chart.js / D3.js – Data visualization for reports

SCREENSHOTS

Blood Bank Management System

Efficiently manage donors, recipients, inventory, and more with our comprehensive blood bank system.

[Register Donor](#)[Register Recipient](#)

Donor Management

Register and manage blood donors

Register new donors and view existing ones



Recipient Management

Register and manage blood recipients

Register new recipients and view existing ones



Blood Inventory

Manage blood stock and inventory

View and update blood inventory levels



Staff Management

Manage blood bank staff

Add and manage staff members



Donations

Record blood donations

Record new blood donations



Transfusions

Record blood transfusions

Record new blood transfusions



Blood Tests

Record blood test results

Record and view blood test results



Reports

Generate and view reports

Access comprehensive blood bank reports



SCREENSHOTS

 **Blood Bank**

[Home](#) [Donors](#) [Recipients](#) [Blood Inventory](#) [Staff](#) [Donations](#) [Transfusions](#) [Blood Tests](#) [Reports](#)

Reports

View and generate blood bank reports

Donors

Recipients

Inventory

 **Donor Report**
List of all registered blood donors

 Export

ID	Name	Age	Gender	Blood Group	Contact
1	Vishal	18	Male	B+	1234567890
2	pranav	18	Male	A+	1234567890
3	sandy	54	Male	O+	0987654321
4	muks	19	Male	B+	9345177558
5	ajr	19	Male	A-	9444597074
6	dmini	18	Female	AB-	6969087654
7	shalini	18	Male	A-	0987552340
8	khYAYHIKA	19	Female	B-	9790800221
9	MEHTA	64	Other	AB+	0987612345
10	uma	33	Female	B+	3344556277

SCREENSHOTS

 **Blood Bank**

Home

Donors

Recipients

Blood Inventory

Staff

Donations

Transfusions

Blood Tests

Reports

Reports

View and generate blood bank reports

Donors

Recipients

Inventory

 **Recipient Report**

List of all registered blood recipients

 Export

ID	Name	Age	Gender	Blood Group	Contact
1	Vishal	25	Male	B+	1234567890
2	vaishnavi s	21	Female	A+	0981234765
3	priyanka	19	Female	A-	6541239870
4	sandyvrox	19	Male	AB+	3214560987
5	harini	18	Female	A-	1234567809
6	mehta	20	Female	AB+	9006377790
7	khushi	20	Female	B+	9790800223
8	madhan	32	Male	O+	9777123453
9	harshika	19	Female	O+	8347921056
10	ammu	18	Female	B+	0651293847
11	sasti	18	Female	A+	7483291056

SCREENSHOTS

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 Blood Bank

Home

Donors

Recipients

Blood Inventory

Staff

Donations

Transfusions

Blood Tests

Reports

Blood Inventory

Manage and monitor blood stock levels

Refresh

Total Units

9

Blood Inventory Status

Current blood stock levels

Blood Group	Quantity (Units)
A+	2
A-	2
O+	1
B+	3
B-	1

SCREENSHOTS

 **Blood Bank**

[Home](#) [Donors](#) [Recipients](#) [Blood Inventory](#) [Staff](#) [Donations](#) [Transfusions](#) [Blood Tests](#) [Reports](#)

Donor Registration

Register a new blood donor in the system

 **Donor Information**

Enter the donor's personal and contact information

Full Name

Age

Gender

Select gender

▼

Blood Group

Select blood group

▼

Contact Number

Must be a 10-digit number.

Register Donor

SCREENSHOTS

Recipient Registration

Register a new blood recipient in the system

Recipient Information

Enter the recipient's personal and contact information

Full Name

John Doe

Age

25

Must be between 18 and 65 years.

Gender

Select gender

Blood Group

Select blood group

Contact Number

1234567890

Must be a 10-digit number.

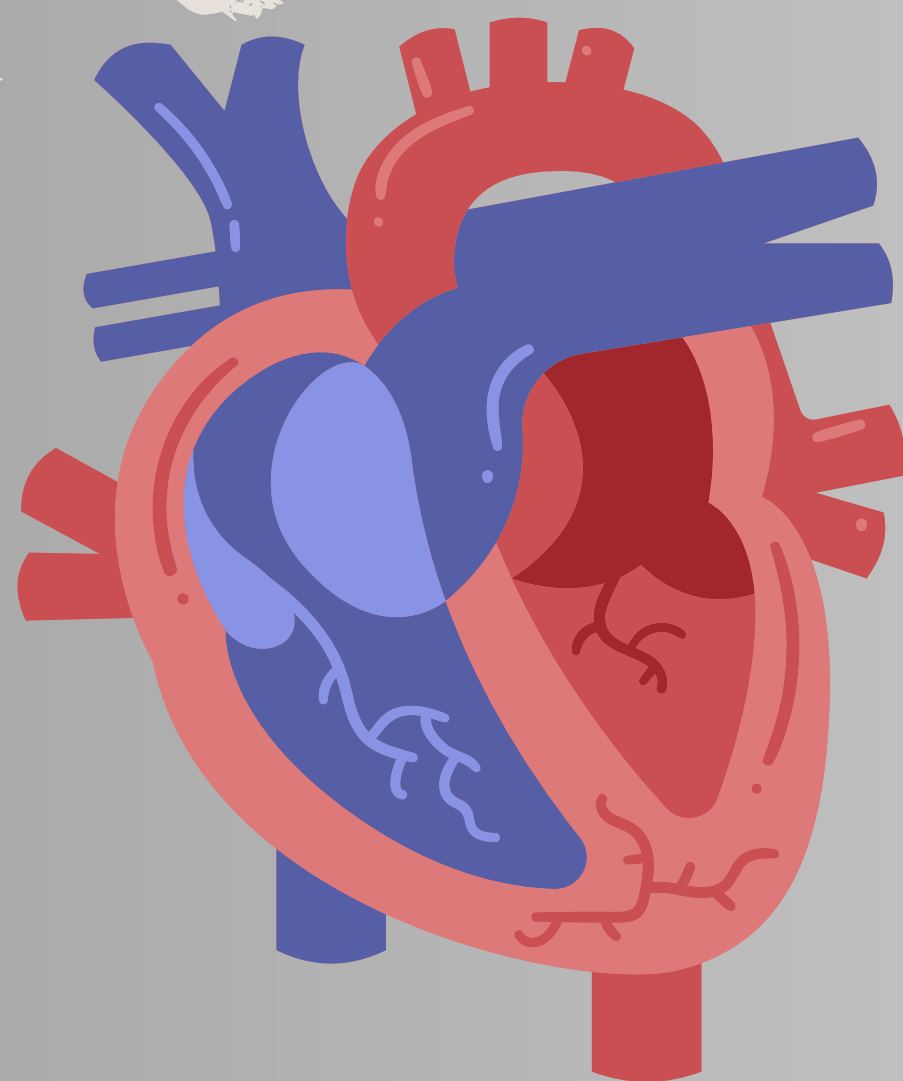
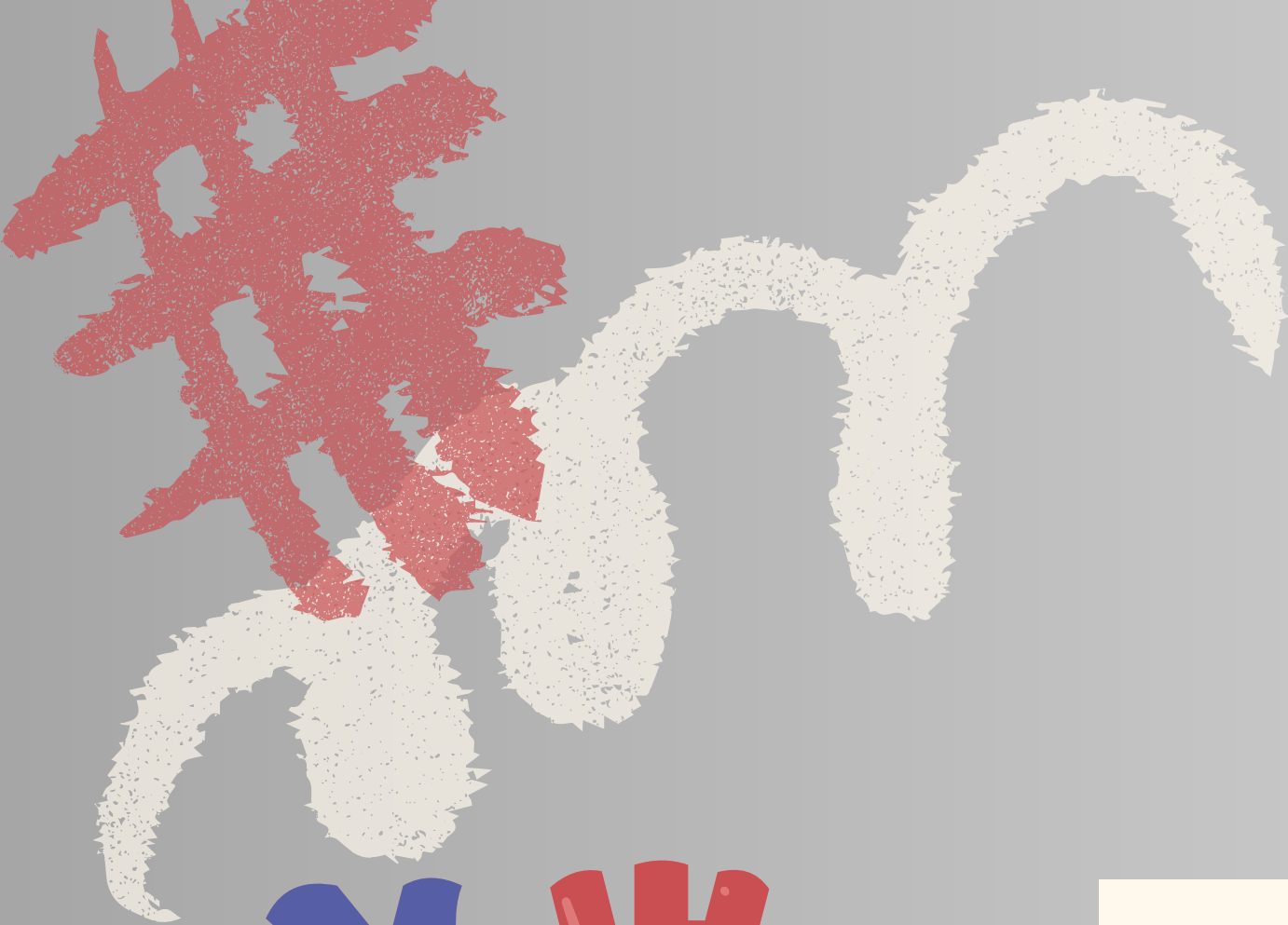
Register Recipient

CONCLUSION

This Blood Bank Management System ensures efficient blood donation, storage, and distribution with accurate records. To make it better and executable, I will:

- Optimize the database for performance.
- Enhance the UI for a seamless experience.
- Implement real-time updates on blood availability.
- Strengthen security for data protection.

These improvements will make the system fully functional and ready for real-world use.



**Thank
You**

